

JOSHUA HIMMENS

Engineering Physics Student

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WORK EXPERIENCE

AltaML

Associate Machine Learning Developer

Jan 2025 – Apr 2025

TRIUMF

ATLAS Deep Learning Research Student

May 2024 – Present

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with Wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting edge transfer learning approaches.
- Implementing models into production using NVIDIA Triton.

UBC Orbit Satellite Team

Command and Data Handling (CDH) Lead and Firmware Developer

Oct 2023 – Present

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of 10 firmware developers, with over 40 tasks completed, 1300 CI builds, 2000 lines of code completed/written.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

Embedded Firmware Developer

Jan 2023 – Present

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computer for launch in Q1 2026.

NumPy

Open Source Contributor

Jan 2021 – Dec 2021

- Contributed to the open-source numerical analysis tool with 86 million monthly downloads.

Elections Alberta

Elections Officer

Jan 2023 – Dec 2023

- Integrated as part of a team to run a safe and fair provincial election.
- Managed public interactions and vote counting.

Western Canada High School

International Baccalaureate Student

Jan 2020 – Dec 2023

- Took the IB diploma program with higher-level Math, Physics, and Chemistry.
- Completed an Extended Essay on simulations of the brachistochrone problem in physics.